

Application Lifecycle Management: Used by the Best of the Best in Class

Business Challenge

IT organizations that are actively engaged in application lifecycle management for their service-oriented architectures (SOA) are more likely to have lowered application service and lifecycle costs than even the Best in Class from Aberdeen's survey for its new *SOA Management and Governance Benchmark Report*. Companies that want to improve how they manage SOA projects must focus on how they develop and maintain services within an SOA. That can lead to the appropriate methodologies and policies, such as code re-use, that drive lower costs of ownership.

Roadmap to Lower Costs and Higher Quality

Between one-third and half of the nearly 1,000 companies Aberdeen surveyed in 2006 regarding their service-oriented architecture (SOA) strategies are having serious difficulties moving their SOA-enabled applications into stable deployment. The predominant reason for these difficulties is inadequate tools for the automated management and governance of the growing plethora and complexity of web services and applications under an SOA, exacerbated by inexperience.

But organizations that have implemented policies on design, governance, and re-use — and that have active processes to manage the application lifecycle — are a step up on the rest, even the Best in Class companies, according to results from a recent benchmark survey for Aberdeen's *SOA Management and Governance Benchmark Report* (Table 1). In fact, these application lifecycle practitioners are all part of the Best in Class group from the survey.

Recommendations for Action

- ✓ Focus on application lifecycle costs.
- ✓ Establish SOA operations, security, governance, and management policies and processes.
- ✓ Plan for the long haul instead of project by project..

Table 1: How Application Lifecycle Management Practitioners Led the Field

	Lifecycle Management Focus	Best in Class	All Respondents
Percent of organizations that decreased SOA development costs over last 12 months	68%	46%	25%
No change in costs	12%	44%	40%
Costs increased	20%	10%	35%

Source: AberdeenGroup, February 2007

The top driver for SOA management and governance is developing new business capabilities or new products and services, followed by managing IT complexity. Standing in the way of optimized SOA operations is establishment of operational security, governance, and management (Table 2).

In response to these challenges, **the top response, chosen by 45% of the survey base, is to revise application lifecycle processes and responsibilities.** This challenge is best illustrated by an automotive parts manufacturer who told us, *“With SOA, we were forced to rewrite whole chapters of our IT processes manual...and that’s not a bad thing.”*

Table 2: Challenges of SOA Operations and How Organizations Respond to Them

Challenges	% Selected	Responses to Challenges	% Selected
Establishment of operational security, governance, and management	45%	Revised our application lifecycle process and responsibilities	45%
Security issues are different from those of older IT	38%	Engaged third-party IT services consulting, evaluation, and assessments	41%
Debugging problems with complex services, composites	37%	Purchase or plan to purchase SOA operations management software	28%
SLAs: Scaling to production volumes, reliability and availability	37%	Purchase or plan to purchase SOA security software	26%
Data management of SOA services is problematic	33%	Invested in additional operations and applications lifecycle training	22%
QA: Testing and deployment stage problems spill over into production	31%	Bought additional computer capacity to minimize performance/capacity issues	19%

Source: [AberdeenGroup](#), February 2007

Management, Governance, and Overall Software Lifecycle Costs

There is a tangible cause-and-effect relationship between management and governance and overall software lifecycle costs:

- Operationally, SOA introduces a myriad of new design, testing, and performance-related issues that can not readily be solved with prior technology or IT elbow grease;
- Design-time governance is where programmers can be “encouraged” to re-use existing services, saving the initial coding time and long-term maintenance costs of writing new but duplicative program code;
- Operational governance ensures that eligible services are given the resources they need, and that performance and security issues are managed.

The inefficiencies caused by a lack of SOA management and governance lower IT performance, raise current operating costs, and collar IT with long-term software maintenance costs that will endure for decades.

The 12% of our survey respondents whose organizations are actively engaged in application lifecycle management for SOA are seeing, on average, a 29% payback from their investments in SOA management tools. In contrast, all organizations represented in the survey have yet to see a positive return (Table 3).

Table 3: Average Investments and Paybacks in SOA Development

	Investment	Payback	ROI
Lifecycle Management Focus	\$2.68 million	\$3.45 million	+29%
All Respondents	\$1.37 million	\$1.18 million	- 14%

Source: [AberdeenGroup](#), February 2007

The Global 1000 Lead the Pack

Given complex, distributed organizations and the IT to support these organizations, the Global 1000 are leading practitioners of application lifecycle management. Our survey found that 48% of the companies focused on application lifecycle management have annual revenue of more than \$1 billion. And 57% have had SOA services deployed for more than a year, compared with 31% of the overall survey. This longevity is evident in 29% having more than 100 SOA services deployed, versus 11% overall.

What's Driving Application Lifecycle Management?

Based on our survey, the companies that focus on application lifecycle management are driven by the following:

- For 42%, SOA governance is an update to existing IT governance processes; and
- Half want to ensure uniform and consistent policy application across the organization;

But application lifecycle management is *not* a standalone process for many. Fifty-two percent have SOA governance processes for BOTH design and run-time operations, compared with 29% of the overall survey and 29% of the Best in Class.

Recommendations for Action

To become more effective at managing SOA strategies, focus on the following:

- **Lower application lifecycle costs.** A systematic focus on the costs of developing, operating, and maintaining SOA services, which can be assisted with automation, will result in the appropriate methodologies and policies such as code re-use that drive lower costs of ownership. Without design-time governance, programmers will not re-use services, and lifecycle costs will balloon. Run-time governance leads to higher quality, uptime, and better security.
- **Establish operational security, governance, and management policies and processes.** “You can’t manage what you can’t measure” is an old, but true, adage. Too many Laggard organizations — those who fell in the bottom 30% of our Competitive Framework in the survey — lack the process, let alone automation for the process. Start with SOA operations and move next to design governance, revising application lifecycle processes and responsibilities along the way.

- **Plan for the long haul.** Too many SOA projects are justified on project ROI under business-as-usual rules. That approach won't unlock the long-term value of lower application lifecycle costs nor solve the management of IT complexity issues driven by a growing use of SOA. Our 2006 surveys showed that companies following an "Enterprise SOA" strategy (using SOA middleware such as an enterprise service bus) can scale more easily with lower short- and long-term costs by investing in all aspects of SOA infrastructure software and in IT staff training.

Related Research

[*The Composite Applications Benchmark Report*](#); December 2006

[*The Legacy Application Modernization Benchmark Report*](#); September 2006

Upcoming Research

The SOA Management and Governance Benchmark Report; coming in March 2007

Locking Down Your SOA: Security Inside and Out; coming in May 2007

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